

Questions :

1. Neural Networks

- (a) Provide an overview of 3 layer feed forward neural network with figure and mathematical notations. [8]
- (b) Explain how multi-layer neural networks can model non-linear decision boundaries while it's basic building block Perceptron(/neuron) is just a linear classifier. [4]
- (c) Briefly explain how learning is performed in neural networks. [4]

2. CNN

- (a) Provide brief description of Convolution operation with formula of 1D Convolution [3+2]
- (b) Explain how CNN's improved upon neural networks while training for high dimensional data such as images. [4]
- (c) The input image to the CNN is of size 29 X 29. It is convoluted with the kernel/ filter of size 5 X 5 with a stride of 1 and zero padding of 1 pixels on each side of an image. Compute the dimension of output feature embedding. [2]
- (d) Briefly explain the concept of *vanishing gradients* in deep learning. [3]

3. Ensemble Learning

- (a) Briefly explain 3 key problems addressed by ensemble learning [3]
- (b) Give short overview of at least two ensemble learning techniques. [4]

[P.T.O.]

4. Data Clustering

[4]

(a) Give visual construction of agglomerative clustering technique.

[4]

(b) Provide psuedo code of Kmeans clustering algorithm.

[5]

(c) List at least 5 limitations of Kmeans clustering algorithm.

5. Time-series Models

(a) Provide a sample state transition probabily matrix and draw associated state transition diagram
(assume at least 3 states in the Markov process). [3+3]

(b) Draw graphical visualization of RNN unfolding with time with appropriate notations for input,
ouput and parameters. [4]

6. Dimensionality Reduction & KNN Classifier

(a) Provide key steps of computing PCA based dimensionality reduction given the input d -dimensional
data matrix $\mathbf{X}$ consisting of n samples. [6]

(b) Provide formula of three similarity measures that can be used in KNN classifier. [3]

7. SVM & Kernel Trick

(a) Give grahpical visualization of example of how kernelization can help in solving non-linear
classification problems. [6]

(b) Provide dual formulation of SVM. [5]

Good Luck !